

FBISE Class 9th Chemistry

Solution 2nd Annual Exam, 2025

Code 10051

Sec A MCQs

1. Which branch studies chemical changes in living organisms?

A — Biochemistry is correct. It studies chemical processes in living things, such as digestion and respiration.

- **B — Analytical chemistry** identifies substances and measures how much of each is present.
- **C — Physical chemistry** studies how and why chemical processes occur, including energy and reaction rates.
- **D — Inorganic chemistry** mainly studies substances such as metals, salts, and minerals.

2. Which group has one electron in its outer shell?

B — Group I is correct. Group I elements have one outer-shell electron. For example, sodium has the electron arrangement **2, 8, 1**.

- **A — Group II:** two outer-shell electrons.
- **C — Group III:** three outer-shell electrons in the group notation used here.
- **D — Group VII:** seven outer-shell electrons.

3. What bond forms when atoms share electron pairs?

B — Covalent bond is correct. In a covalent bond, atoms *share* one or more pairs of electrons.

- **A — Ionic bond:** electrons are transferred from one atom to another.
- **C — Metallic bond:** metal atoms are held together by attraction involving mobile electrons.
- **D — Hydrogen bond:** an attraction involving hydrogen and nearby electronegative atoms; it is not the bond formed by sharing electron pairs in this question.

4. Which substance is a saturated hydrocarbon?

D — $\text{CH}_3\text{—CH}_3$ is correct. This is ethane. It contains only carbon and hydrogen, and its carbon atoms are joined by a **single bond**. That makes it a saturated hydrocarbon.

- **A — $\text{CH}_2=\text{CH}_2$:** contains a carbon–carbon *double* bond, so it is unsaturated.
- **B — $\text{CH}_2=\text{CHCl}$:** contains a double bond, so it is unsaturated. It also contains chlorine, so it is not a hydrocarbon.
- **C — $\text{CH}\equiv\text{CH}$:** contains a carbon–carbon *triple* bond, so it is unsaturated.

5. What is the reducing agent in $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$?

B — CO is correct. Carbon monoxide removes oxygen from iron oxide, turning Fe_2O_3 into Fe. A substance that causes another substance to be reduced is called a *reducing agent*. CO itself gains oxygen and becomes CO_2 .

- **A — Fe_2O_3 :** is being reduced; it is the **oxidizing agent** in this reaction.
- **C — Fe:** is a product formed after iron oxide is reduced.
- **D — CO_2 :** is the product formed after CO is oxidized.

Easy way to remember Q5: CO takes oxygen from Fe_2O_3 , so CO is the reducing agent.

6. Electrons in higher energy levels are ...

D — Far from the nucleus. Think of energy levels as layers around the nucleus: a higher level is generally farther out.

- **A, closer to the nucleus:** describes lower energy levels.

- **B, less energetic:** higher levels have *more* energy.
- **C, stable:** “stable” does not describe their position; electrons in higher levels can also move to lower levels.

7. Number of atoms in 2 moles of O₂

None of the printed options is correct for *atoms*. One O₂ molecule has **two oxygen atoms**:

$$\left[2 \times \text{mol O}_2 \times 6.022 \times 10^{23} \times \frac{\text{molecules}}{\text{mol}} \times 2 \times \frac{\text{atoms}}{\text{molecule}} \right] = \boxed{2.4088 \times 10^{24} \text{ atoms}}$$

- **A — 3.01×10^{23} :** too small.
- **B — 6.022×10^{23} :** the number of *molecules* in **one** mole of O₂.
- **C — 1.204×10^{24} :** approximately the number of *molecules* in **two** moles of O₂, but the question asks for atoms.
- **D — 2.0×10^{23} :** too small.

For teaching or marking: If the intended word was “**molecules**,” C would be the answer. With “**atoms**” as printed, the MCQ needs correction.

8. Which ion is usually formed by chlorine?

C — Cl⁻. A chlorine atom usually **gains one electron**. An electron has a negative charge, so the resulting chloride ion has a charge of -1.

- **A — Cl⁺:** would mean chlorine lost one electron.
- **B — Cl²⁻:** would mean it gained two electrons; that is not its usual ion.
- **D — Cl²⁺:** would mean it lost two electrons.

9. Functional group in amines

D — -NH₂. This is the **amino group** shown in a simple primary amine, such as CH₃NH₂.

- **A — -OH:** characteristic of alcohols.

- **B — —COOH :** characteristic of carboxylic acids.
- **C — —CHO :** characteristic of aldehydes.

10. One main source of protein

A — Meat. Meat contains substantial protein, which the body uses to build and repair tissues.

- **B — Oil:** mainly fat.
- **C — Water:** contains no protein.
- **D — Sugar:** a carbohydrate, not a protein.

11. Catalytic converters help control emissions of ...

C — CO and NO. A catalytic converter in a vehicle helps change harmful **carbon monoxide (CO)** and **nitrogen oxides (including NO)** into less harmful substances.

- **A — Cl_2 :** chlorine gas is not the exhaust pollutant targeted here.
- **B — N_2 , O_2 :** nitrogen and oxygen are ordinary major components of air.
- **D — He:** helium is not a vehicle exhaust pollutant.

12. Most compressible state of matter

A — Gas. Gas particles have large spaces between them. Pressure can push them much closer together.

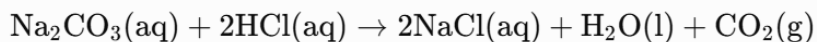
- **B — Liquid:** its particles are already fairly close together.
- **C — Solid:** its particles are tightly packed.
- **D — Plasma:** can also be compressible, but **gas is the expected Class 9 answer** to this standard comparison.

Sec B Short Questions

(i) Sodium carbonate and hydrochloric acid

Gas evolved: carbon dioxide (CO₂). When an acid reacts with a carbonate, carbon dioxide gas is released. The bubbles observed are this gas.

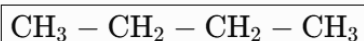
Balanced equation:



The upward bubbles can be confirmed as CO₂ by passing the gas through limewater, which turns milky.

OR — Structure and classification of C₄H₁₀

One structural formula for C₄H₁₀ is **butane**:



It is a **saturated hydrocarbon** because it contains only carbon and hydrogen, and **all carbon-carbon bonds are single bonds**. Its molecular formula also follows the alkane formula C_nH_{2n+2} : for $n = 4$, $H = 10$.

(ii) Relative charge and relative mass

Particle	Relative charge	Relative mass
Electron	-1	Approximately 1/1836
Proton	+1	1
Neutron	0	1

The electron has a very small mass compared with a proton or neutron; the neutron has **no electric charge**.

OR — Three differences between metals and non-metals

Property	Metals	Non-metals
Heat and electrical conduction	Generally good conductors	Generally poor conductors
Malleability	Can usually be hammered into sheets	Solid non-metals are usually brittle
Electron behaviour in reactions	Usually lose electrons and form positive ions	Often gain electrons and form negative ions

These are **general trends**, so exceptions exist. For example, graphite is a non-metal that conducts electricity.

there are other states such as Bose-Einstein Condensates(BEC) which is defined as the state of matter in which separate atoms cooled to temperatures very close to absolute zero. BEC is observable under extreme conditions of cold temperature. Superfluid and superconductors are the two main materials which contain BEC.

OR — Oxidation numbers in polyatomic ions

Rule: The sum of the oxidation numbers of all atoms in a polyatomic ion equals the overall charge of that ion.

a. Ammonium, NH_4^+ : Let nitrogen's oxidation number be x . Hydrogen is usually $+1$.

$$x + 4(+1) = +1 \Rightarrow \boxed{x = -3}$$

Check: $-3 + 4 = +1$, equal to the ion's charge.

b. Carbonate, CO_3^{2-} : Let carbon's oxidation number be x . Oxygen is usually -2 .

$$x + 3(-2) = -2 \Rightarrow \boxed{x = +4}$$

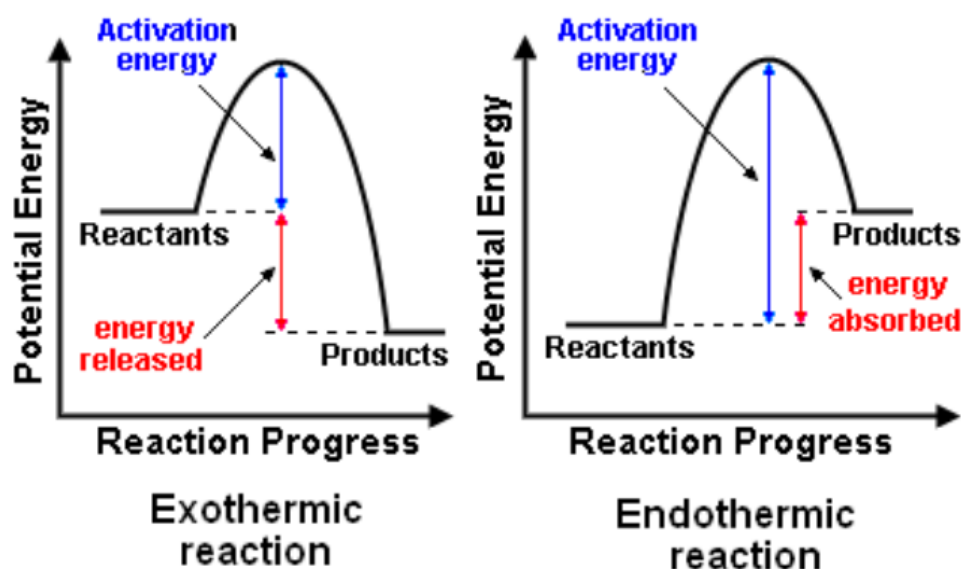
Check: $+4 - 6 = -2$, equal to the ion's charge.

(iv) Greenhouse effect

The **greenhouse effect** is the warming of Earth's surface and lower atmosphere when certain gases absorb some of the infrared energy emitted by Earth and send energy back in different directions, including towards the surface.

Any two greenhouse gases: carbon dioxide (CO_2) and methane (CH_4). Water vapour is another valid example.

v)



v) Nitrates and phosphates in water

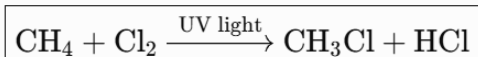
Sources: Nitrates and phosphates enter water mainly through runoff from **fertilized fields, animal manure, and sewage or wastewater**. Some detergents can also add phosphates to wastewater.

Why harmful: Excess nitrate and phosphate encourage rapid growth of algae. When the algae die and decay, dissolved oxygen in the water falls, which can harm or kill fish. This process is called **eutrophication**. High nitrate levels in *drinking water* can also be especially dangerous for infants because they can reduce the blood's ability to carry oxygen.

OR — Substitution reaction and halogenation of methane

A **substitution reaction** is one in which an atom or group in a molecule is **replaced** by another atom or group.

In sunlight or ultraviolet light, one hydrogen atom in methane can be replaced by a chlorine atom:



Here, CH_3Cl is chloromethane. One H in CH_4 has been substituted by Cl. Further substitution is possible if chlorination continues, but this equation gives the required first step.

(vi) Halogens and trends down the group

Halogens are the elements of **Group VII (Group 17)** of the periodic table, including fluorine, chlorine, bromine and iodine.

Property

Trend down the group

Density

Generally increases

Reactivity

Decreases

Halogens react by gaining an electron. Down the group, the outer shell is farther from the nucleus and is more shielded by inner electrons, so gaining an electron becomes less easy.

OR — Complete the disease table

Disease	Cause	Common symptoms
Dysentery	Infection acquired through contaminated food or water; <i>Shigella</i> bacteria are one cause	Diarrhoea, sometimes bloody, with abdominal pain or cramps and fever
Typhoid	<i>Salmonella Typhi</i> bacteria, commonly spread through contaminated food or water	Persistent fever, headache, abdominal pain; diarrhoea or constipation may occur
Hepatitis	In this waterborne-disease context, usually hepatitis A virus from contaminated food or water	Tiredness, nausea, dark urine and jaundice (yellow eyes or skin)

(vii) Aldehydes and ketones

Both contain a **carbonyl group**, $C=O$. Their difference is **where that group is placed**:

	Aldehyde	Ketone
General structure	$R-CHO$	$R-CO-R'$
Carbonyl position	At the end of the carbon chain	Within the carbon chain
Example	Ethanal, CH_3-CHO	Propanone, $CH_3-CO-CH_3$

In an aldehyde, the carbonyl carbon is bonded to **at least one hydrogen**. In a ketone, it is bonded to **two carbon-containing groups**.

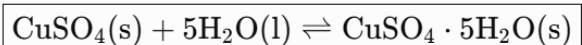
OR — Group I and Group VII valence electrons and ions

Group	Valence electrons	Usual change	Ion formed	Example
Group I	1	Loses one electron	+1 cation	$Na \rightarrow Na^+ + e^-$
Group VII	7	Gains one electron	-1 anion	$Cl + e^- \rightarrow Cl^-$

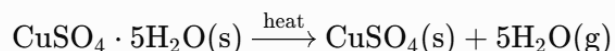
Memory aid: Group I has **one to lose**; Group VII needs **one to complete eight**.

(viii) Adding water to anhydrous copper(II) sulfate

Observation: Anhydrous copper(II) sulfate, CuSO_4 , is **white**. On adding water, it turns **blue** as hydrated copper(II) sulfate forms.



White anhydrous solid + water $\rightleftharpoons$ blue hydrated solid. On heating the blue crystals, water is driven off and the solid becomes white again:



The question calls the observed colour change a “physical change.” For a precise explanation, **the visible observation is a change from white to blue; its cause is hydration and formation of the hydrated compound.**

OR — One source of each nutrient

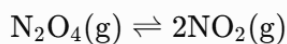
Nutrient	One suitable food source
Carbohydrate	Rice
Protein	Eggs
Lipids (fats)	Cooking oil

(ix) Chemical equilibrium

Chemical equilibrium is the state in a reversible reaction at which the **forward and reverse reactions occur at equal rates**. The amounts of reactants and products then remain constant, although both reactions continue.

Conditions: The reaction must be **reversible** and take place in a **closed system**, so substances cannot escape or enter. Under constant conditions, such as temperature, the forward and reverse rates can become equal.

For example:



At equilibrium, $\text{rate}_{\text{forward}} = \text{rate}_{\text{reverse}}$. **Equal rates do not mean equal amounts** of reactants and products.

OR — Strong acid versus weak acid

	Strong acid	Weak acid
Dissociation in water	Almost completely dissociates into ions	Partially dissociates; many acid molecules remain undissociated
Example	Hydrochloric acid, HCl	Ethanoic acid, CH_3COOH
Equation	$\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$	$\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{CH}_3\text{COO}^-$

Exam point: *Strength* means the **extent of ionization**, not how concentrated the acid solution is. A dilute HCl solution is still a solution of a strong acid.

(x) Mole and mole calculations

A **mole** is an amount of substance containing 6.022×10^{23} specified particles, such as atoms or molecules. Use:

$$n = \frac{m}{M}$$

where n is moles, m is mass in grams, and M is molar mass in g mol^{-1} .

a. 48 g of ozone, O_3

$$M(\text{O}_3) = 3(16) = 48 \text{ g mol}^{-1} \quad n = \frac{48}{48} = \boxed{1 \text{ mol}}$$

b. 14 g of nitrogen gas, N_2

$$M(\text{N}_2) = 2(14) = 28 \text{ g mol}^{-1} \quad n = \frac{14}{28} = \boxed{0.5 \text{ mol}}$$

OR — Reaction in respiration

Cellular respiration involves the **oxidation of glucose** and is **exothermic**: it releases energy that cells can use.



Thus, glucose reacts with oxygen to form carbon dioxide and water while releasing energy. In cells, this occurs through a series of enzyme-controlled steps, rather than one single step.

(xi) Relative atomic mass from isotope abundances

The isotopes shown have mass numbers 12, 13 and 14, with printed abundances 98.8%, 1.1% and 0.009%. All three have atomic number 6, as isotopes of the same element must.

Using the weighted-average formula:

$$\begin{aligned} A_r &= \frac{(12 \times 98.8) + (13 \times 1.1) + (14 \times 0.009)}{100} \\ &= \frac{1185.6 + 14.3 + 0.126}{100} \\ &= \boxed{12.00026 \approx 12.0} \end{aligned}$$

OR — Organic versus inorganic chemistry

Organic chemistry

Studies **carbon compounds**, particularly those with carbon–hydrogen bonds, and their reactions.

Example: **methane** (CH₄) or **ethanol** (C₂H₅OH).

Inorganic chemistry

Studies compounds generally outside the organic category, including metals, salts, minerals and acids.

Example: **sodium chloride** (NaCl) or **sulfuric acid** (H₂SO₄).

Sec C Long Questions

Q3 Electronic configuration and position in the periodic table

Given configuration: $1s^2 2s^2 2p^6 3s^1$. Its 11 electrons give atomic number 11, so element X is sodium (Na). Its outer shell has one electron, placing it in Group I (Group 1). Three occupied main shells place it in Period 3.

Property	Sodium Na	Potassium K below Na
Electron arrangement	2, 8, 1	2, 8, 8, 1
Common ion	Na^+	K^+
Atomic size	Smaller	Larger
Reactivity	Less reactive	More reactive

Both are alkali metals with one valence electron and similar reactions.

Potassium has an additional shell, greater shielding and a weaker attraction for its outer electron. It loses that electron more easily and is more reactive.

Q3 OR Dot and cross diagrams for sodium fluoride and water

In the figure, red dots are electrons from fluorine or oxygen; blue crosses are electrons from sodium or hydrogen.

Q3 (OR): Dot-and-cross electron diagrams

Red dots = electrons originally from F or O; blue crosses = electrons originally from Na or H.

(i) Sodium fluoride (NaF) — ionic bond



Na⁺ has lost its outer electron; F⁻ has eight outer electrons (7 dots + 1 cross).

(ii) Water (H₂O) — two covalent bonds



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Sodium fluoride: Na transfers its one outer electron to F. Na⁺ and F⁻ are held together by electrostatic attraction, forming an ionic bond. Fluorine now has eight outer electrons (seven dots and one cross).



Water: Oxygen shares one electron with each hydrogen. Each O–H shared pair is a single covalent bond. Oxygen retains two lone pairs; each H achieves a duet and oxygen completes an octet.

Q4 Arrhenius and Brønsted Lowry concepts

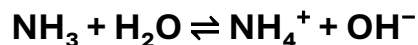
Concept	Acid	Base
Arrhenius	Produces H ⁺ (H ₃ O ⁺) in water	Produces OH ⁻ in water
Brønsted–Lowry	Donates a proton H ⁺	Accepts a proton H ⁺

Arrhenius examples: HCl in water produces H₃O⁺, whereas NaOH in water produces OH⁻.



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Brønsted–Lowry example: NH_3 accepts a proton from H_2O and is a base; H_2O donates that proton and acts as an acid. The proton-transfer concept is broader because NH_3 does not contain OH^- in its formula.



Q4 OR Alkanes and their preparation

Alkanes are saturated hydrocarbons with only C–C single bonds; their general formula is $\text{C}_n\text{H}_{2n+2}$. They are relatively unreactive because their C–C and C–H bonds are strong and they have no reactive functional group. Under suitable conditions, they do undergo combustion and substitution.

Method 1 Hydrogenation of alkenes. Hydrogen adds across a double bond in the presence of a nickel catalyst:



Method 2 Decarboxylation of sodium carboxylates. Heating sodium ethanoate with soda lime produces methane:



A further valid method is the Wurtz reaction: $2\text{CH}_3\text{Cl} + 2\text{Na} \xrightarrow{\text{dry ether}} \text{CH}_3-\text{CH}_3 + 2\text{NaCl}$.

Q5 Moles from particle counts

One mole contains $N_a = 6.022 \times 10^{23}$ specified particles. For each part, use $n = N/N_a$.

Part	Substitution	Answer
a CO_2 molecules	$1.2044 \times 10^{24} \div 6.022 \times 10^{23}$	2.0 mol
b Ti atoms	$3.011 \times 10^{23} \div 6.022 \times 10^{23}$	0.5 mol
c CH_2O molecules	$3.011 \times 10^{22} \div 6.022 \times 10^{23}$	0.05 mol

d Zn atoms	$1.2044 \times 10^{23} \div 6.022 \times 10^{23}$	0.2 mol
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CO₂ and CH₂O were counted as molecules; Ti and Zn were counted as atoms. Do not multiply by the number of atoms within a molecule unless the question asks for atoms.

Q5 OR Oxidizing and reducing agents

Oxidation is an increase in oxidation number; reduction is a decrease. The oxidized substance is the reducing agent; the reduced substance is the oxidizing agent.

Reaction a: $2S + Cl_2 \rightarrow S_2Cl_2$. In the product, chlorine is -1 and sulfur is $+1$ per atom ($2S + 2[-1] = 0$).

Element	Oxidation number change	Process
S	$0 \rightarrow +1$	Oxidation
Cl	$0 \rightarrow -1$	Reduction

Therefore, S is the reducing agent and Cl₂ is the oxidizing agent.

Reaction b: $WO_3 + 3H_2 \rightarrow W + 3H_2O$. In WO₃, W is $+6$ because each O is -2 . Elemental W and H₂ are both 0; H becomes $+1$ in water.

Element	Oxidation number change	Process
W	$+6 \rightarrow 0$	Reduction
H	$0 \rightarrow +1$	Oxidation

Therefore, WO₃ is the oxidizing agent and H₂ is the reducing agent.

Q6 Comparison of solutions and colloids

Feature	True solution	Colloid
Nature	Homogeneous	Appears uniform but is heterogeneous at particle level
Particle size	Generally below 1 nm	Approximately 1 to 1000 nm
Light scattering	No Tyndall effect	Shows Tyndall effect
Example	Salt dissolved in water	Milk

The Tyndall effect means that the path of a narrow light beam becomes visible as dispersed particles scatter light. Any three comparison points plus one example of each satisfy the question.

Q6 OR Biomolecules and their functions

Biomolecules are molecules found in living organisms that build the body or take part in life processes.

Biomolecule	One major function
Carbohydrates	Provide readily available energy
Proteins	Build and repair tissues; many enzymes are proteins
Lipids	Store energy over the long term and form cell membranes

Nucleic acids such as DNA are another valid type; they store and transmit genetic information.